

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 20%

QUESTIONS:3

Total Marks:30

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- Analyze all the given constraints and explain how they affect the robot's decision-making process.
- Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I P POORNA CHANDU(192412199)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5

Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6
Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

ANSWER 1:

Given

Doctors: D1,D2,D3

Shifts:Morning (M),Afternoon (A),Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3. (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor must be assigned exactly one shift.

Step 1: Start

No doctor is assigned.

Doctor	Shift
D1	-
D2	-
D3	-

Step 2: Assign D1

Morning

Constraint Check

D1 is not assigned Night, One doctor per shift.

Doctor	Shift
D1	Morning
D2	-
D3	-

Step 3: Assign D2

Morning already assigned to D1.

Try Afternoon

Constraint Check

Unique shift,Current Assignment.

Doctor	Shift
D1	Morning
D2	Afternoon
D3	-

Step 4: Assign D3

D3 cannot work Morning.

Afternoon already occupied by D2.

Try Night

Constraint Check

D3 not Morning.

Afternoon < Night

Therefore

D2 works before D3 , Only one doctor per shift, Every doctor assigned exactly one shift.

Doctor	Shift
D1	Morning
D2	Afternoon
D3	Night

ANSWER 2:

Given Constraints

- Grid size = 5×5
- Allowed moves = **Up, Down, Left, Right**
- Cost of each move = **1**
- Robot cannot move through obstacles
- Manhattan Distance heuristic:

$$h(n) = |x_g - x_n| + |y_g - y_n| \quad h(n) = |x_g - x_n| + |y_g - y_n| \quad h(n) = |x_g - x_n| + |y_g - y_n|$$

Grid Representation

- S = Start = (0,0)
- G = Goal = (4,4)
- X = Obstacle

Step 1: Initialize

Visited = {(0,0)}

Queue:

Step 2: Expand (0,0)

- Right $\rightarrow$ (0,1)
- Down $\rightarrow$ (1,0)

Step 3: Expand (0,1)

- Right $\rightarrow$ (0,2)
- Down $\rightarrow$ (1,1) Obstacle

Step 4: Expand (1,0)

- Down $\rightarrow$ (2,0)
- Right $\rightarrow$ (1,1) Obstacle

Step 5: Expand (0,2)

- Down $\rightarrow$ (1,2)
- Right $\rightarrow$ (0,3) Obstacle

Step 6: Expand (2,0)

- Down $\rightarrow$ (3,0)
- Right $\rightarrow$ (2,1) Obstacle

Step 7: Expand (1,2)

- Down $\rightarrow$ (2,2)

Step 8: Expand (3,0)

- Right $\rightarrow$ (3,1)

Step 9: Expand (2,2)

- Right $\rightarrow$ (2,3)

Step 10: Expand (3,1)

- Down $\rightarrow$ (4,1)

Step 11: Expand (2,3)

- Right $\rightarrow$ (2,4)

Step 12: Expand (4,1)

- Right $\rightarrow$ (4,2)

Step 13: Expand (4,2)

- Right $\rightarrow$ (4,3)

Step 14: Expand (4,3)

- Right $\rightarrow$ Goal (4,4) ✓

Goal reached.

Cost Calculation:

Number of moves = 8

Cost per move = 1

Total Cost = $8 \times 1 = 8$ $\text{Total Cost} = 8 \times 1 = 8$

Final Answer

Total Cost: 8 units

ANSWER 3:

Given Constraints

- Robot can move Up, Down, Left, Right only.
- Obstacles (X) cannot be crossed.
- Robot can observe only adjacent cells.
- Cost of each movement = 1.
- Entering a risky zone (R) adds an extra cost of 2.
- Robot must update its knowledge dynamically (Online Search).
- Objective: Reach the goal with the minimum cost

Grid Representation

- S = Start
- G = Goal
- X = Obstacle
- R = Risky Zone
- 0 = Safe Cell

Step 1: Start

Robot starts at S (0,0).

Current Cost = 0

Step 2: Expand Start Node

The robot observes adjacent cells.

- Right $\rightarrow (0,1)$, Cost = 1
- Down $\rightarrow (1,0)$, Cost = 1

Both have equal cost. Select (1,0).

Step 3: Expand (1,0)

Adjacent cells:

- Down $\rightarrow (2,0)$, Cost = 2
- Right $\rightarrow (1,1)$ = Obstacle

Expand the lowest-cost node.

Step 4: Expand (0,1)

Adjacent cells:

- Right $\rightarrow (0,2)$ = Risk Zone

Cost = 1 + 3 = 4

- Down $\rightarrow$ Obstacle

Priority Queue

Robot chooses (2,0) because it has the lower cost.

Step 5: Expand (2,0)

Possible move:

Priority Queue

Step 6: Expand (2,1)

Possible move:

Priority Queue

Choose (2,2).

Step 7: Expand (2,2)

Possible moves:

- Right $\rightarrow (2,3)$ Risk Zone

Cost = 7

- Down $\rightarrow (3,2)$

Cost = 5

Priority Queue

Robot avoids the risky zone and selects **(3,2)**.

Step 8: Expand (3,2)

Possible move:

Priority Queue

Step 9: Expand (4,2)

Possible move:

Priority Queue

Step 10: Reach Goal

From **(4,3)**

Move to

Total Cost Calculation

Move	Cost
8 Normal Moves	8
Risk Zones Entered	0
Total Cost	8

a) Constraint Analysis

Constraint	Effect
Up, Down, Left, Right only	Robot moves in four directions only.

Obstacles	Robot cannot cross blocked cells.
Limited sensing	Robot observes only adjacent cells.
Movement cost = 1	Each move increases total cost by 1.
Risk zone = +2	Robot prefers safer paths to reduce cost.
Online search	Robot updates its map while moving.

(b) Solution Approach (Uniform Cost Search - UCS)

1. Start from S.
2. Observe adjacent cells.
3. Add valid neighboring cells to the priority queue.
4. Expand the node with the **lowest cumulative cost**.
5. Avoid obstacles and risky zones whenever possible.
6. Update the map if new information is found.
7. Repeat until the robot reaches G.

(c) AI Concepts

1. Intelligent Agent

The robot senses the environment, makes decisions, and moves toward the goal.

2. Rationality

The robot chooses the **minimum-cost and safest path**.

3. Environment Type

Environment	Description
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Partially Observable	Only adjacent cells are visible.
Dynamic	New information is discovered while moving.
Sequential	Each move affects the next decision.
Discrete	The robot moves one cell at a time.

(d) Justification

Uniform Cost Search (UCS) is suitable because it:

- Finds the **minimum-cost path**.
- Considers movement and risk costs.
- Updates decisions dynamically.
- Avoids obstacles and risky zones efficiently.